

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\evan\Documents\GitHub\hw3-evtw`
Warning: The project dependencies or compat requirements have changed since the manifest was generated.
It is recommended to `Pkg.resolve()` or consider `Pkg.update()` if necessary.
@ Pkg.API C:\Users\evan\.julia\juliaup\julia-1.12.1+0.x64.w64.mingw32\share\julia\stdlib\v1.12
Precompiling packages...
```

MbedTLS

HTTP

0 dependencies successfully precompiled in 7 seconds. 211 already precompiled.

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

Streeter-Phelps solution:

$$U \frac{dC}{dx} = k_a(C_s - C) + P - R - S_B - k_c B_0 \exp\left(\frac{-k_c x}{U}\right) - k_n N_0 \exp\left(\frac{-k_n x}{U}\right)$$

In this model, we ignore P, R, S_B .

$$U \frac{dC}{dx} = k_a(C_s - C) - k_c B_0 \exp\left(\frac{-k_c x}{U}\right) - k_n N_0 \exp\left(\frac{-k_n x}{U}\right)$$

From the known variables:

$$k_a = \frac{0.55}{day}$$

$$k_c = \frac{0.35}{day}$$

$$k_n = \frac{0.25}{day}$$

$$U = \frac{6km}{day}$$

$$C_s = 10 \frac{mg}{L}$$

Need to split the system into two parts: the stream before Stream 2 is integrated and the river after Stream 2 is added:

Integrating the equation:

$$C(x) = C_s(1 - \alpha_1) + C_0 \alpha_1 - B_0 \alpha_2 - N_0 \alpha_3$$

$$\alpha_1 = \exp\left(-\frac{k_a x}{U}\right)$$

$$\alpha_2 = \left(\frac{k_c}{k_a - k_c}\right) \left[\exp\left(-\frac{k_c x}{U}\right) - \exp\left(-\frac{k_a x}{U}\right)\right]$$

$$\alpha_3 = \left(\frac{k_n}{k_a - k_n}\right) \left[\exp\left(-\frac{k_n x}{U}\right) - \exp\left(-\frac{k_a x}{U}\right)\right]$$

Need to find the initial concentration C_0 when $x = 0$. Let $x = 0$ when the river and Stream 1 just mix:

$$\alpha_1 = e^{-0} = 1$$

$$\alpha_2 = \left(\frac{k_c}{k_a - k_c}\right)(1 - 1) = 0$$

$$\alpha_3 = \left(\frac{k_n}{k_a - k_n}\right)(1 - 1) = 0$$

$$C(x) = C_0$$

Initial concentration of C_0, B_0, N_0 :

$$\begin{aligned}
C_0 &= 7.5 \frac{mg}{L} \times \frac{10}{11} + 5 \frac{mg}{L} \times \frac{1}{11} \approx 7.27 \frac{mg}{L} \\
B_0 &= 5 \frac{mg}{L} \times \frac{10}{11} + 50 \frac{mg}{L} \times \frac{1}{11} \approx 9.09 \frac{mg}{L} \\
N_0 &= 5 \frac{mg}{L} \times \frac{10}{11} + 35 \frac{mg}{L} \times \frac{1}{11} \approx 7.73 \frac{mg}{L}
\end{aligned}$$

Next, we need to consider the system after Stream 2 is added when $x = 15\text{km}$. First, the concentration of the river before Stream 2 is added must be calculated:

$$\begin{aligned}
C(x = 15) &= C_s(1 - \alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3 \\
\alpha_1 &= e^{-\frac{0.55 \times 15}{6}} \approx 0.252 \\
\alpha_2 &= \frac{0.35}{0.55 - 0.35} (e^{-\frac{0.35 \times 15}{6}} - e^{-\frac{0.55 \times 15}{6}}) \approx 0.287 \\
\alpha_3 &= \frac{0.25}{0.55 - 0.25} (e^{-\frac{0.25 \times 15}{6}} - e^{-\frac{0.55 \times 15}{6}}) \approx 0.235
\end{aligned}$$

Calculating the concentration of the river before Stream 2 is added:

$$C(x = 15) = 10(1 - 0.252) + 7.27(0.252) - 9.09(0.287) - 7.73(0.235) \approx 4.89 \frac{mg}{L}$$

New system equation after Stream 2 is added:

$$\begin{aligned}
C(x) &= C_s(1 - \alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3 \\
\alpha_1 &= \exp\left(-\frac{k_a(x - 15)}{U}\right) \\
\alpha_2 &= \left(\frac{k_c}{k_a - k_c}\right) \left[\exp\left(-\frac{k_c(x - 15)}{U}\right) - \exp\left(-\frac{k_a(x - 15)}{U}\right)\right] \\
\alpha_3 &= \left(\frac{k_n}{k_a - k_n}\right) \left[\exp\left(-\frac{k_n(x - 15)}{U}\right) - \exp\left(-\frac{k_a(x - 15)}{U}\right)\right]
\end{aligned}$$

We subtract x by 15 as the mixing starts at $x = 15$. The initial concentrations after the river and Stream 2 need to be calculated:

$$\begin{aligned}
C_0 &= 4.89 \frac{mg}{L} \times \frac{110}{125} + 5 \frac{mg}{L} \times \frac{15}{125} \approx 4.90 \frac{mg}{L} \\
B_0 &= 9.09(1 - 0.287) \frac{mg}{L} \times \frac{110}{125} + 45 \frac{mg}{L} \times \frac{15}{125} \approx 11.10 \frac{mg}{L} \\
N_0 &= 7.73(1 - 0.235) \frac{mg}{L} \times \frac{110}{125} + 35 \frac{mg}{L} \times \frac{15}{125} \approx 9.40 \frac{mg}{L}
\end{aligned}$$

With these two system equations, it is possible to plot the dissolved oxygen concentration over distance downstream using a for loop and continually generating values of DO down the river. The minimum value in the stream is printed below:

```

# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.27 # mg/L
B1_0 = 9.09 # mg/L
N1_0 = 7.73 # mg/L

# System 2:
C2_0 = 4.90 # mg/L
B2_0 = 11.10 # mg/L
N2_0 = 9.40 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

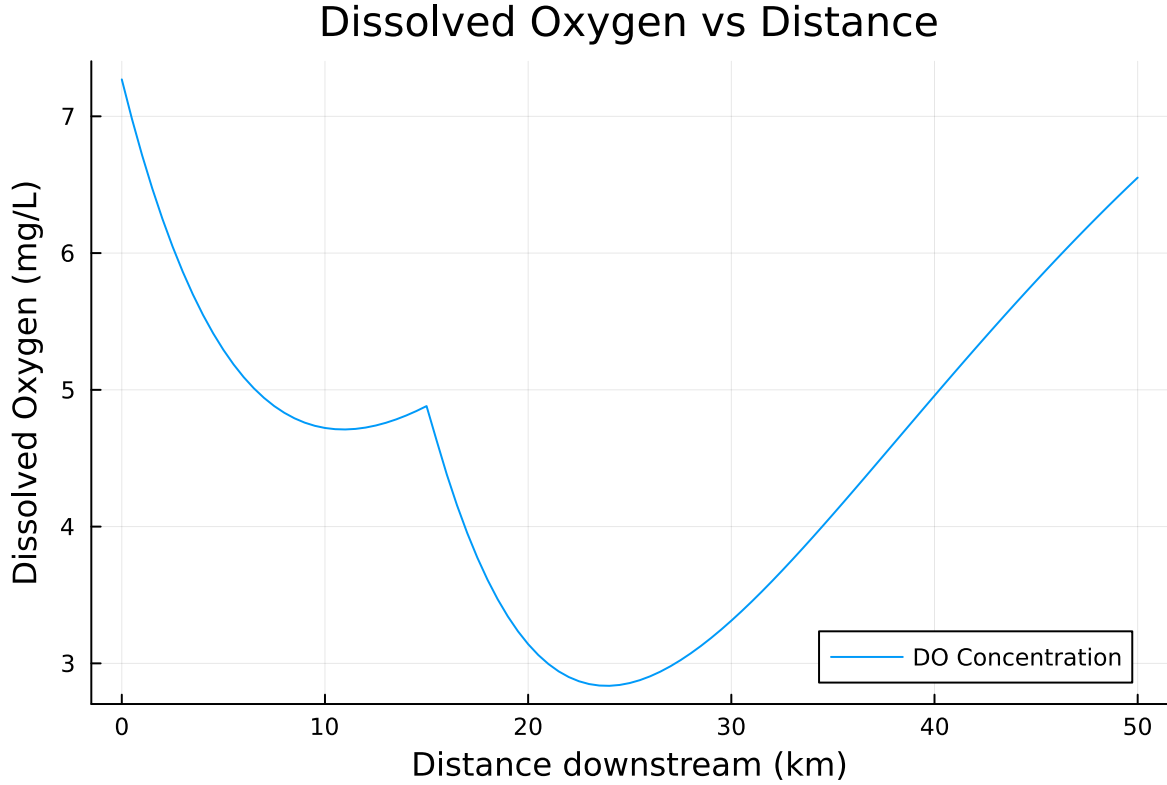
        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

println("Minimum DO value in stream = ",minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen")

```

Minimum DO value in stream = 2.8358280713478363



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km.

Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

Numerical integrated solution will be utilizing the Forward Euler discretization.

$$U \frac{dC}{dx} = k_a(C_s - C) - k_c B_0 \exp\left(\frac{-k_c x}{U}\right) - k_n N_0 \exp\left(\frac{-k_n x}{U}\right)$$

$$\frac{dC}{dx} = \frac{k_a}{U}(C_s - C) - \frac{k_c}{U} B_0 \exp\left(\frac{-k_c x}{U}\right) - \frac{k_n}{U} N_0 \exp\left(\frac{-k_n x}{U}\right)$$

$$As \frac{dC}{dx} \approx \frac{\Delta C(x)}{\Delta x} = \frac{C(x + \Delta t) - C(x)}{\Delta x}$$

$$\frac{C(x + \Delta t) - C(x)}{\Delta x} = \frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 \exp\left(\frac{-k_c x}{U}\right) - \frac{k_n}{U}N_0 \exp\left(\frac{-k_n x}{U}\right)$$

$$C(x + \Delta t) - C(x) = \Delta x \left[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 \exp\left(\frac{-k_c x}{U}\right) - \frac{k_n}{U}N_0 \exp\left(\frac{-k_n x}{U}\right) \right]$$

$$C(x + \Delta t) = C(x) + \Delta x \left[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 \exp\left(\frac{-k_c x}{U}\right) - \frac{k_n}{U}N_0 \exp\left(\frac{-k_n x}{U}\right) \right]$$

For $t = 0.5$,

$$C(x + 0.5) = C(x) + 0.5 \left[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 \exp\left(\frac{-k_c x}{U}\right) - \frac{k_n}{U}N_0 \exp\left(\frac{-k_n x}{U}\right) \right]$$

When Stream 2 is added ($x \geq 15$), the equation will change:

$$C_0 = C(x = 15) * \frac{110}{125} + 5 * \frac{15}{125}$$

$$B_0 = B(x = 15) * \frac{110}{125} + 45 * \frac{15}{125}$$

$$N_0 = N(x = 15) * \frac{110}{125} + 45 * \frac{15}{125}$$

$$C(x + \Delta t) = C(x) + \Delta x \left[\frac{k_a}{U}(C_s - C(x)) - \frac{k_c}{U}B_0 \exp\left(\frac{-k_c(x - 15)}{U}\right) - \frac{k_n}{U}N_0 \exp\left(\frac{-k_n(x - 15)}{U}\right) \right]$$

By creating functions to simulate the values of the timestep and concentration, the numerical solution can be plotted and compared to the analytical solution. Two separate timestep simulation functions are used depending on the distance downstream.

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

function simulate_timestep1(C, i, C0, B0, N0)
    return k_a/U*(C_s - C) - k_c/U*B0*exp(-k_c*i/U) - k_n/U*N0*exp(-k_n*i/U)
end
```

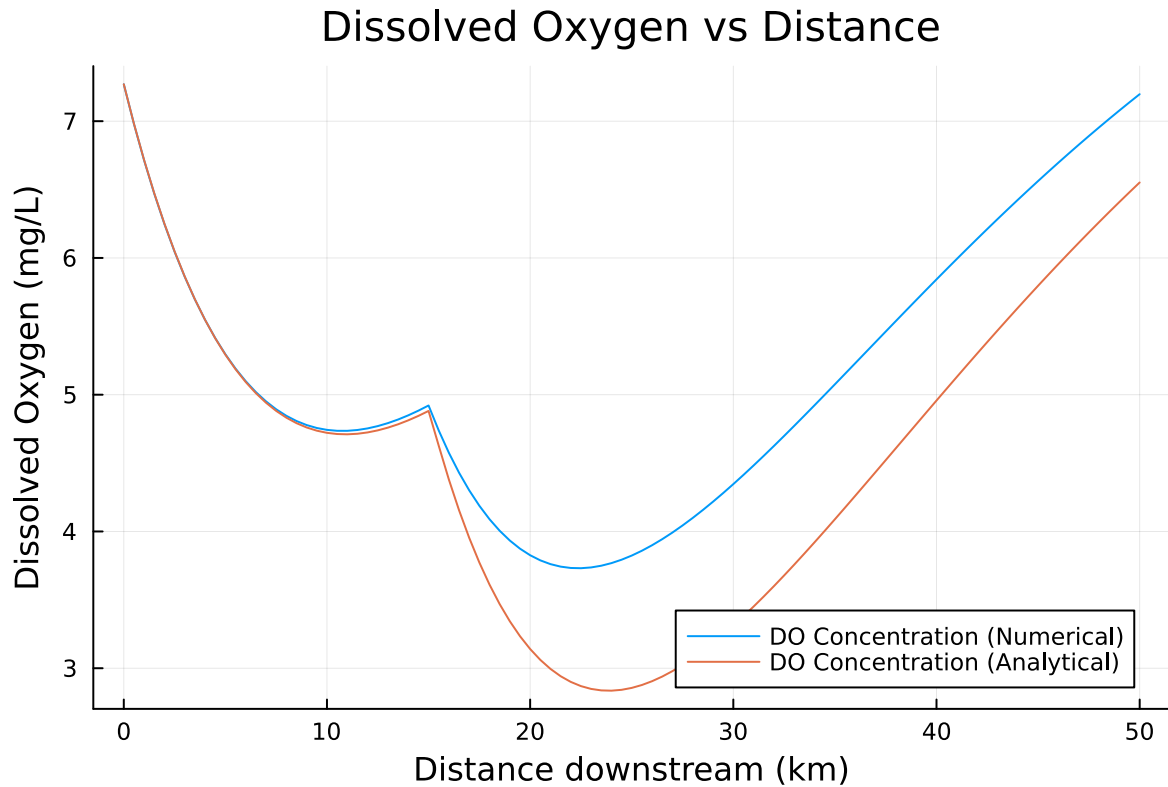
```

function simulate_timestep2(C, i, C0, B0, N0)
    return k_a/U*(C_s - C) - k_c/U*B0*exp(-k_c*(i-15)/U) - k_n/U*N0*exp(-k_n*(i-15)/U)
end

function simulate_wastewater(C0, B0, N0, distance, dx)
    steps = Int64(distance / dx)
    C = zeros(steps + 1)
    C[1] = C0
    for i in 1:steps
        if i*dx <= 15
            C[i+1] = C[i] + dx * simulate_timestep1(C[i], i*dx, C0, B0, N0)
        else
            if i*dx == 15
                C0 = C[i]*110/125 + 5*15/125
                B0 = (B0 - B0*exp(-k_c*i/U))*110/125 + 45*15/125
                N0 = (N0 - N0*exp(-k_c*i/U))*110/125 + 45*15/125
            end
            C[i+1] = C[i] + dx * simulate_timestep2(C[i], i*dx, C0, B0, N0)
        end
    end
    return C
end

plot(distance_range, simulate_wastewater(7.27, 9.09, 7.73, 50, 0.5), xlabel="Distance downstream", ylabel="DO Concentration (Numerical)", legend=true)
plot!(distance_range, DO_values, label="DO Concentration (Analytical)", legend=true)

```

Caption: Comparison of DO content modelled analytically and numerically along the river from 0 to 50km.

From this plot, the DO concentration plotted numerically is significantly different than the plot analytically. The values of DO plotted numerically after Stream 2 is added is significantly larger than the DO values plotted analytically along the same domain. This difference can be explained by the timestep in the numerical solution accounting for changes in DO concentrations, whereas the analytical solution assumes that changes in concentration is based on its initial values. Since the rate of decay would be based on the current concentration, the numerical solution would have a lower rate of change in DO than the analytical solution, therefore having larger DO values along the river.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Since the streams are being treated, the initial concentrations C_0, B_0, N_0 for both control volumes before Stream 2 is added and after need to be changed. Provided that there is a 5% margin of safety, the absolute limit is 3.8 mg/L. ($4 \text{ mg/L} \times 0.95 = 3.8 \text{ mg/L}$)

We'll start with the scenario when waste is treated by 50%. We'll also assume that the total mass inflow of the treated streams do not change.

Initial concentration of C_0, B_0, N_0 before Stream 2 is added:

$$C_0 = 7.5 \frac{\text{mg}}{\text{L}} \times \frac{10}{11} + 5 \frac{\text{mg}}{\text{L}} \times 0.5 \times \frac{1}{11} \approx 7.05 \frac{\text{mg}}{\text{L}}$$

$$B_0 = 5 \frac{\text{mg}}{\text{L}} \times \frac{10}{11} + 50 \frac{\text{mg}}{\text{L}} \times 0.5 \times \frac{1}{11} \approx 6.82 \frac{\text{mg}}{\text{L}}$$

$$N_0 = 5 \frac{\text{mg}}{\text{L}} \times \frac{10}{11} + 35 \frac{\text{mg}}{\text{L}} \times 0.5 \times \frac{1}{11} \approx 6.14 \frac{\text{mg}}{\text{L}}$$

$$C(x) = C_s(1 - \alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$

$$\alpha_1 = e^{-\frac{0.55 \times 15}{6}} \approx 0.252$$

$$\alpha_2 = \frac{0.35}{0.55 - 0.35} (e^{-\frac{0.35 \times 15}{6}} - e^{-\frac{0.55 \times 15}{6}}) \approx 0.287$$

$$\alpha_3 = \frac{0.25}{0.55 - 0.25} (e^{-\frac{0.25 \times 15}{6}} - e^{-\frac{0.55 \times 15}{6}}) \approx 0.235$$

Concentration at $x = 15$:

$$C(x = 15) = 10(1 - 0.252) + 7.05(0.252) - 6.82(0.287) - 6.14(0.235) \approx 5.85 \frac{\text{mg}}{\text{L}}$$

Next, the initial concentrations of C_0, B_0, N_0 after Stream 2 is calculated, assuming that the composition of Stream 2 is not changed:

$$C_0 = 5.85 \frac{\text{mg}}{\text{L}} \times \frac{110}{125} + 5 \frac{\text{mg}}{\text{L}} \times \frac{15}{125} \approx 5.75 \frac{\text{mg}}{\text{L}}$$

$$B_0 = 6.82(1 - 0.287) \frac{\text{mg}}{\text{L}} \times \frac{110}{125} + 45 \frac{\text{mg}}{\text{L}} \times \frac{15}{125} \approx 9.68 \frac{\text{mg}}{\text{L}}$$

$$N_0 = 6.14(1 - 0.235) \frac{\text{mg}}{\text{L}} \times \frac{110}{125} + 35 \frac{\text{mg}}{\text{L}} \times 0.5 \times \frac{15}{125} \approx 8.79 \frac{\text{mg}}{\text{L}}$$

After, the same code as in Problem 1.1 is used to plot DO concentration over the stream, except the initial concentrations can be modified by changing the multiplier in Stream 1. A red line at DO = 3.8 mg/L is added to check if the plot satisfies the safety limit.

```

# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 0.5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 0.5 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 0.5 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

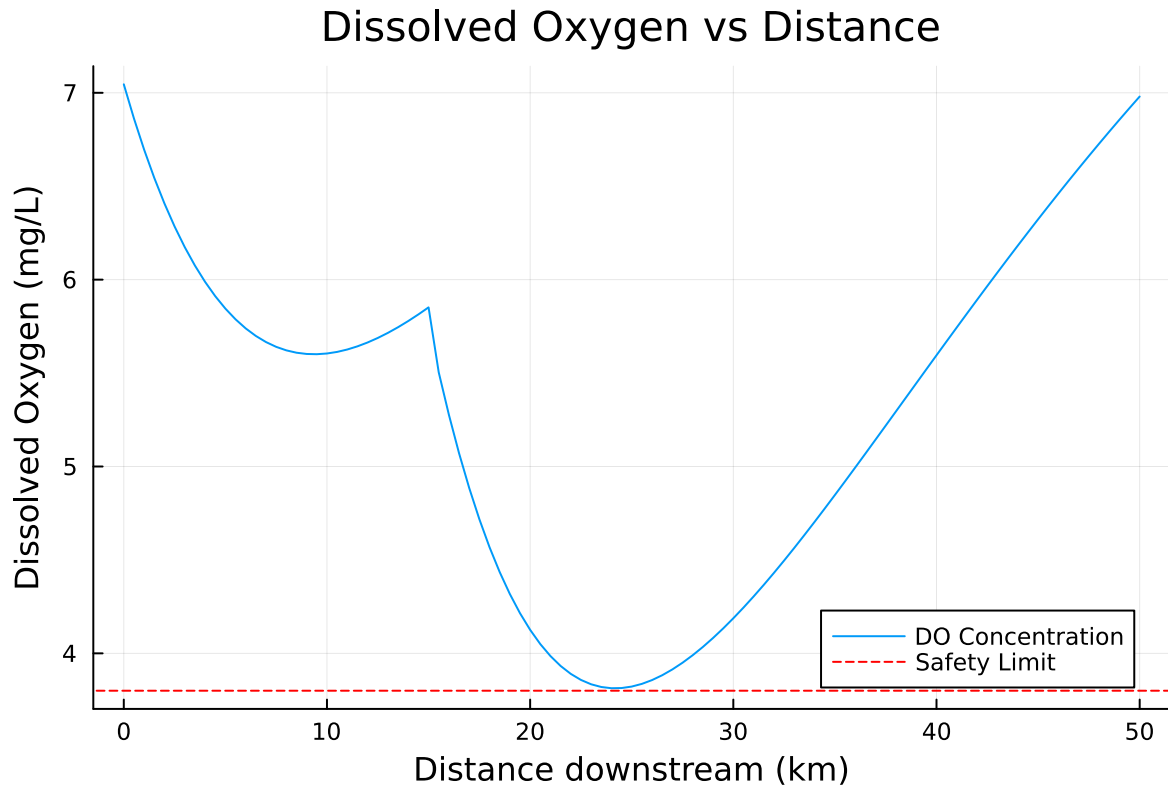
        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println("Minimum DO value in stream = ", minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen

```

```
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

Minimum DO value in stream = 3.8133967134609676



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km with 50% waste treated from Stream 1.

Since this example fortunately skimmed the absolute safety limit in the first try, only one additional test is necessary by confirming if slightly decreasing the treatment % to 49% will make the system still meet the safety limits.

```
# Equation constants:
```

```
k_a = 0.55
```

```
k_c = 0.35
```

```
k_n = 0.25
```

```
C_s = 10 # mg/L
```

```
U = 6 # km/d
```

```
# System 1:
```

```

C1_0 = 7.5 * 10/11 + 5 * 0.51 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 0.51 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 0.51 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

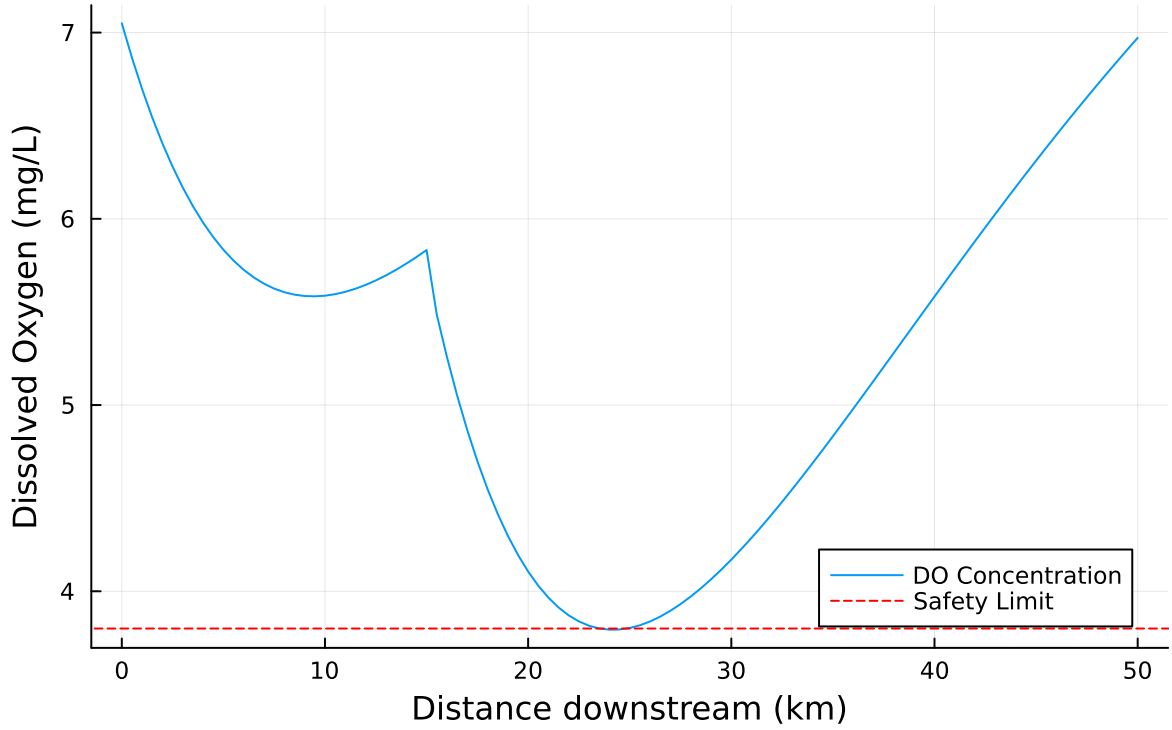
        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println("Minimum DO value in stream = ", minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen")
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)

```

Minimum DO value in stream = 3.7938260915969124

Dissolved Oxygen vs Distance



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km with 49% waste treated from Stream 1.

As decreasing the % removal of waste to 49% does not meet the absolute safety limit, therefore the minimum percentage of waste to be removed in Stream 1 must be at 50% to meet safety limits.

For the scenario where only Stream 2 is treated, the initial concentrations C_0, B_0, N_0 and the concentration at $x = 15$ will be the same as Problem 1.1:

$$C_0 = 7.5 \frac{mg}{L} \times \frac{10}{11} + 5 \frac{mg}{L} \times \frac{1}{11} \approx 7.27 \frac{mg}{L}$$

$$B_0 = 5 \frac{mg}{L} \times \frac{10}{11} + 50 \frac{mg}{L} \times \frac{1}{11} \approx 9.09 \frac{mg}{L}$$

$$N_0 = 5 \frac{mg}{L} \times \frac{10}{11} + 35 \frac{mg}{L} \times \frac{1}{11} \approx 7.73 \frac{mg}{L}$$

Concentration at $x = 15$:

$$C(x = 15) = 10(1 - 0.252) + 7.05(0.252) - 6.82(0.287) - 6.14(0.235) \approx 5.85 \frac{mg}{L}$$

Initial concentrations C_0, B_0, N_0 after Stream 2 is mixed in will be different based on the amount treated. As a starter, a reduction of 50% will be conducted:

$$C_0 = 5.85 \frac{mg}{L} \times \frac{110}{125} + 5 \frac{mg}{L} \times \frac{15}{125} \times 0.5$$

$$B_0 = 9.09(1 - 0.287) \frac{mg}{L} \times \frac{110}{125} + 45 \frac{mg}{L} \times \frac{15}{125} \times 0.5$$

$$N_0 = 7.73(1 - 0.235) \frac{mg}{L} \times \frac{110}{125} + 35 \frac{mg}{L} \times \frac{15}{125} \times 0.5$$

```
# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 # mg/L

# System 2:
C2_0 = (10*(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.5 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.5 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 * 0.5 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))
```

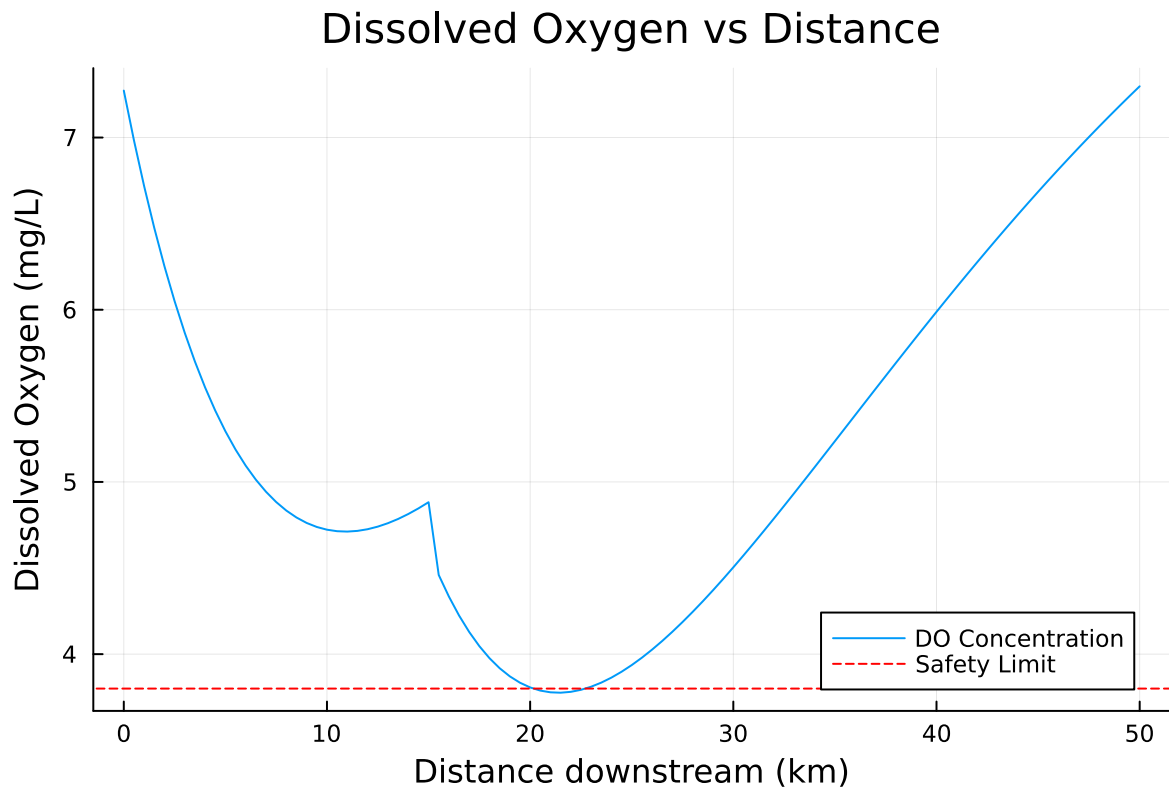
```

    C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
    push!(DO_values, C)
end
end

minimum(DO_values)
println("Minimum DO value in stream = ", minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)

```

Minimum DO value in stream = 3.7760111801567486



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km with 50% waste treated from Stream 2.

From this simulation, it appears that the system only drops below the safety limit for a small period. Therefore, an additional trial will be conducted where the minimum waste reduced is increased by 52%.


```

# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.48 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.48 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 * 0.48 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

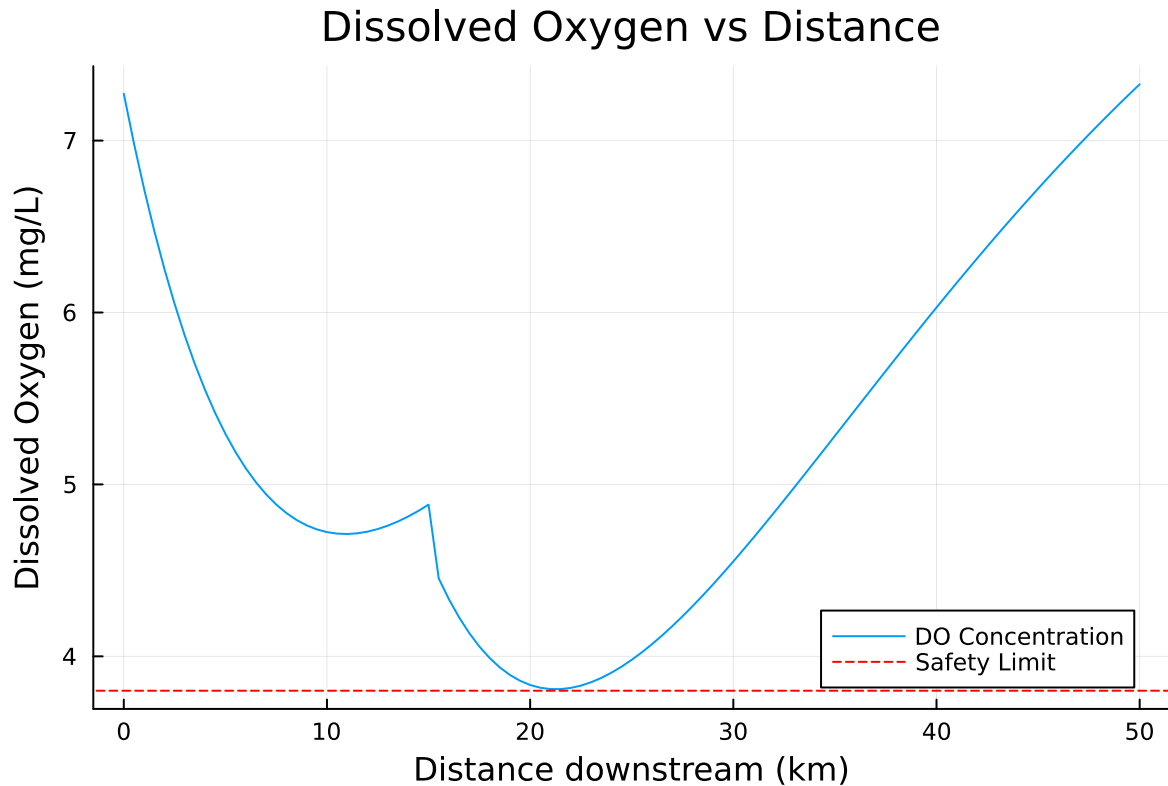
        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println("Minimum DO value in stream = ", minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen")

```

```
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

Minimum DO value in stream = 3.809411846219514



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km with 52% waste treated from Stream 2.

As this amount of waste treated meets the absolute safety limit, therefore the minimum waste needed to be treated from Stream 2 is 52%.

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

To evaluate which option to opt for, a separate simulation is conducted where 25% of waste is removed from both Stream 1 and Stream 2.

```

# Equation constants:
k_a = 0.55
k_c = 0.35
k_n = 0.25
C_s = 10 # mg/L
U = 6 # km/d

# System 1:
C1_0 = 7.5 * 10/11 + 5 * 1/11 * 0.75 # mg/L
B1_0 = 5 * 10/11 + 50 * 1/11 * 0.75 # mg/L
N1_0 = 5 * 10/11 + 35 * 1/11 * 0.75 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 * 0.75 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 * 0.75 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 * 0.75 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

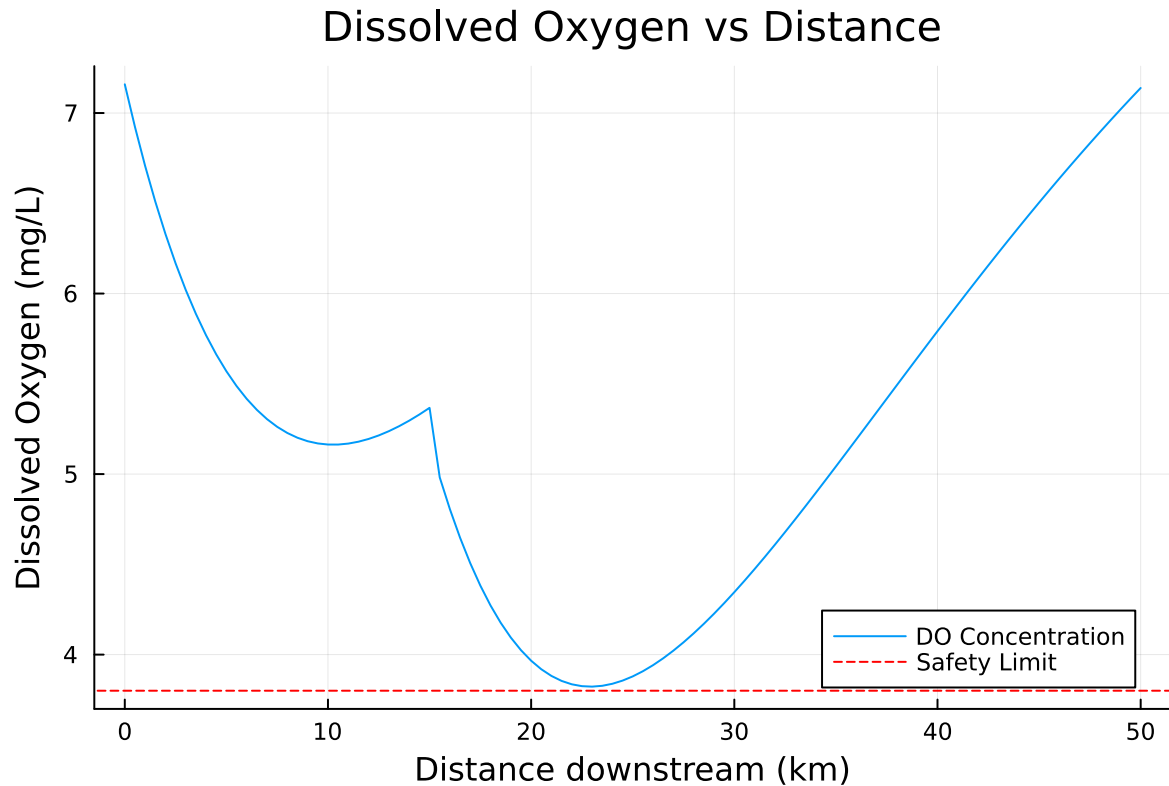
        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end

minimum(DO_values)
println("Minimum DO value in stream = ", minimum(DO_values))
plot(distance_range, DO_values, xlabel="Distance downstream (km)", ylabel="Dissolved Oxygen")

```

```
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)
```

Minimum DO value in stream = 3.8226341788166613



Caption: Line graph of downstream distance along the river vs dissolved oxygen content from 0 to 50km with 25% waste treated from both Stream 1 and Stream 2.

Based on the simulations, it is preferable to treat Stream 2 alone. Since the percentage of waste removed from either stream has similar effects on the magnitude and distance that the minimum value occurs, other considerations about how DO changes through the river should be analyzed. One consideration is the difference in magnitude that DO concentration changes when Stream 2 is added when only Stream 2 is treated vs both streams. When Stream 1 is treated, the DO concentration remains at higher levels from 0km to 15km, but drops significantly after Stream 2 is added. This drastic change in DO could be harmful as it could affect the behaviours of aquatic organisms at that point in the river, even if DO eventually increases later downstream.

Some other information that may be a factor in making a conclusion could be financial and energy costs of treating each stream. Since the effects of removing the same percentage of

waste from each stream is similar, if one method of treatment ends up costing less than the other, then that method would be preferred.

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

For this problem, the same code in Problem 1.3 is used to generate DO values down the river, except the initial concentration $C1_0$ is randomly generated according to a LogNormal distribution. Additionally, instead of plotting DO concentration over distance, the minimum DO value for each random $C1_0$ is recorded and the average minimum DO is plotted over the number of Monte Carlo situations used. A 95% confidence interval will also be recorded in this plot and a separate cumulative distribution curve is generated to find the expected probability. In these simulations, the assumption is made that the percentage of waste treated in Stream 1 is 50% based on Problem 1.3.

To determine if the Monte Carlo sample size is large enough, the Monte Carlo Standard Error (Standard deviation of all values divided by the number of samples) should be less than $5e-5$. While the error tolerance can be modified depending on how much accuracy is desired, $5e-5$ was selected as a compromise between reasonable accuracy and minimizing computational time. To be able to reliably confirm and reproduce the same results, the seed is fixed before any random numbers are generated.

```
Random.seed!(1647)

DO_dist = LogNormal(2.0,0.15)
sample_size = 2000
DO_samples = rand(DO_dist, sample_size)
final_DO_values = [];

for i in 1:sample_size
    # Equation constants:
    k_a = 0.55
    k_c = 0.35
    k_n = 0.25
    C_s = 10 # mg/L
    U = 6 # km/d
```

```

# System 1:
C1_0 = DO_samples[i] * 10/11 + 5 * 0.5 * 1/11 # mg/L
B1_0 = 5 * 10/11 + 50 * 0.5 * 1/11 # mg/L
N1_0 = 5 * 10/11 + 35 * 0.5 * 1/11 # mg/L

# System 2:
C2_0 = (10(1-0.252) + C1_0*0.252 - B1_0*0.287 - N1_0*0.235) * 110/125 + 5 * 15/125 # mg/L
B2_0 = B1_0*(1 - 0.287) * 110/125 + 45 * 15/125 # mg/L
N2_0 = N1_0*(1 - 0.235) * 110/125 + 35 * 15/125 # mg/L

distance_range = [0:0.5:50;];
DO_values = [];

for d in distance_range
    if d <= 15
        alpha1 = exp(-k_a * d / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d / U) - exp(-k_a * d / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d / U) - exp(-k_a * d / U))

        C = C_s*(1-alpha1) + C1_0*alpha1 - B1_0*alpha2 - N1_0*alpha3
        push!(DO_values, C)
    else
        d_adjusted = d - 15
        alpha1 = exp(-k_a * d_adjusted / U)
        alpha2 = (k_c / (k_a - k_c)) * (exp(-k_c * d_adjusted / U) - exp(-k_a * d_adjusted / U))
        alpha3 = (k_n / (k_a - k_n)) * (exp(-k_n * d_adjusted / U) - exp(-k_a * d_adjusted / U))

        C = C_s*(1-alpha1) + C2_0*alpha1 - B2_0*alpha2 - N2_0*alpha3
        push!(DO_values, C)
    end
end
push!(final_DO_values, minimum(DO_values))
end

# print(final_DO_values)
avg_DO = zeros(length(final_DO_values))
std_DO = zeros(length(final_DO_values))
std_DO[1] = 0.0
for j in 1:length(final_DO_values)
    avg_DO[j] = mean(final_DO_values[1:j])
    std_DO[j] = std(final_DO_values[1:j]) / sqrt(j)
end

```

```

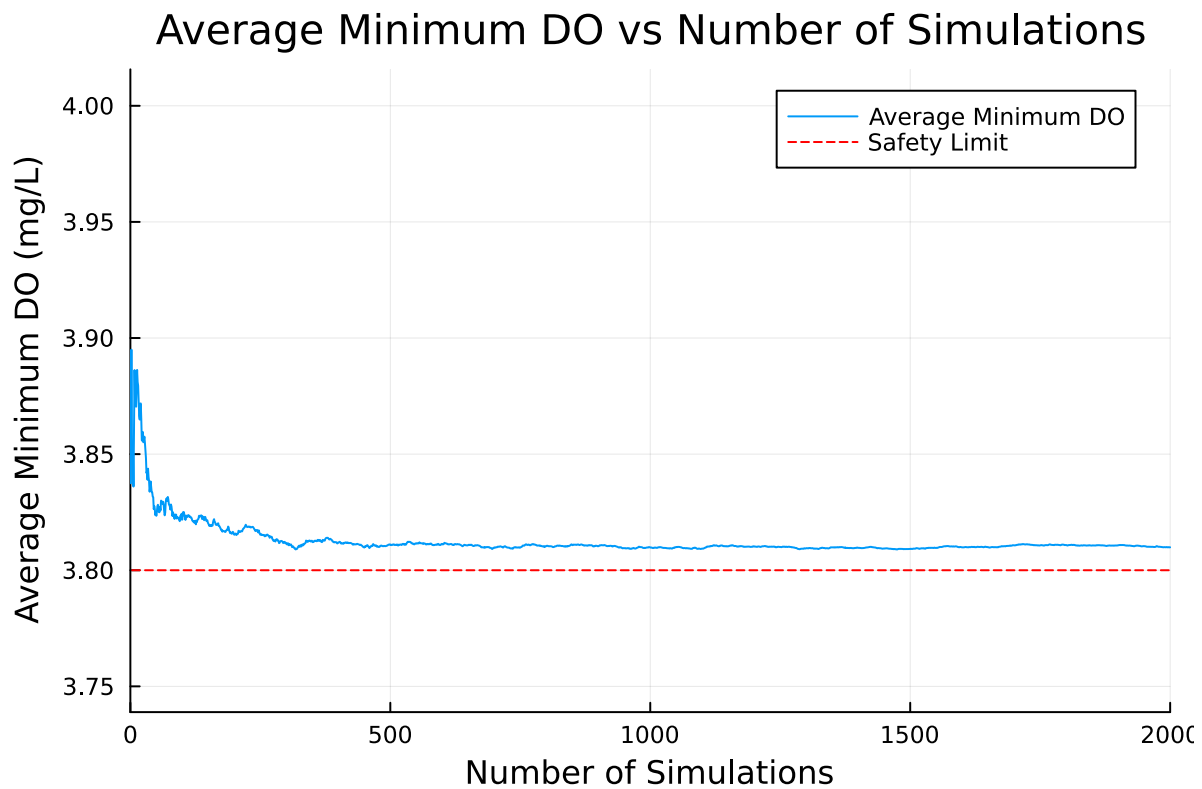
MCSE = std_DO[end] / sqrt(length(final_DO_values))
println("Monte Carlo Standard Error = ", MCSE)
println("Average Minimum DO = ", avg_DO[end])
println("STD of Average Minimum DO = ", std_DO[end])
plot(1:length(final_DO_values), avg_DO, ribbon=1.96 * std_DO, xlim=(0, length(final_DO_values))
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)

```

Monte Carlo Standard Error = 4.860716858799314e-5

Average Minimum DO = 3.8098803888782324

STD of Average Minimum DO = 0.0021737786631309025



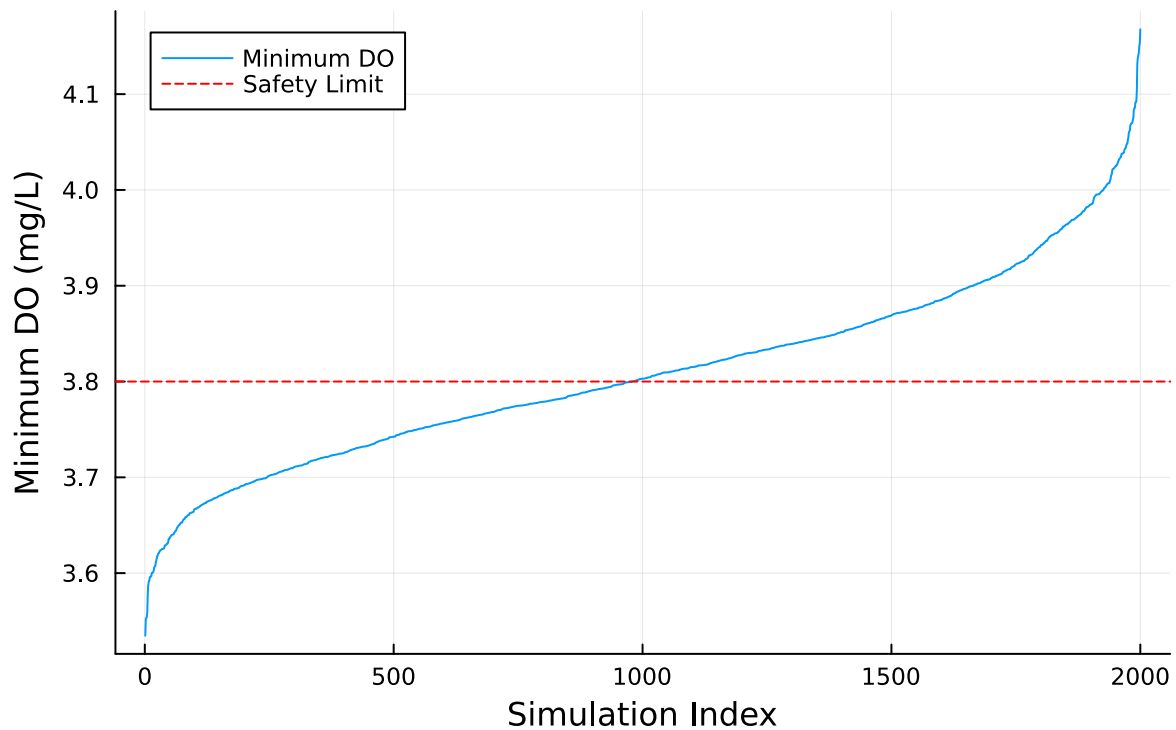
Caption: Monte Carlo simulation graph comparing the average minimum DO from separate cases over the number of samples used. Shaded area = 95% confidence interval.

```

sorted_DO = sort(final_DO_values)
plot(sorted_DO, xlabel="Simulation Index", ylabel="Minimum DO (mg/L)", title="Sorted Minimum
hline!([3.8], label="Safety Limit", linestyle=:dash, color=:red)

```

Sorted Minimum DO from Simulations



Caption: Cumulative distribution curve of the number of simulations used vs the minimum DO content.

Based on these two plots, the expected probability that the river complies with the safety limit is around 50%, since the 3.8 mg/L limit line intersects at around simulation number 1000 out of 2000. Based on the confidence intervals plotted on the number of simulations vs average minimum DO, it should be expected that 95% of samples should be between 3.808 and 3.812.

References

List any external references consulted, including classmates.

Week 4 to 6 Lecture Slides.

AI has been used to cross examine answers.